

JOINT SET AND OPERATIONS

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STEP-1 CREATE DATABASE

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the 'Server Enterprise' tree with 'Schemas' selected. The main pane shows the 'Query Editor' with the following SQL script:

```
1 CREATE DATABASE IF NOT EXISTS joins_db;  
2 USE joins_db;
```

The right pane shows a message: "Automatic context help is disabled. Use the toolbar to manual caret position or to toggle automatic h".

The bottom pane shows the 'Output' window with the following results:

#	Time	Action	Message
1	19:47:43	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'joins_db'; database exists
2	19:47:43	USE joins_db	0 row(s) affected

STEP-2 CREATE TABLES

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the 'Server Enterprise' tree with 'Schemas' selected. The main pane shows the 'Query Editor' with the following SQL script:

```
1 CREATE DATABASE IF NOT EXISTS joins_db;  
2 USE joins_db;  
3 CREATE TABLE class (  
4     id INT,  
5     name VARCHAR(30)  
6 );  
7  
8 CREATE TABLE class_info (  
9     id INT,  
10    address VARCHAR(30)  
11 );
```

The right pane shows a message: "Automatic context help is disabled. Use the toolbar to manual caret position or to toggle automatic h".

The bottom pane shows the 'Output' window with the following results:

#	Time	Action	Message
1	19:47:43	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'joins_db'; database exists
2	19:47:43	USE joins_db	0 row(s) affected
3	19:49:03	CREATE TABLE class (id INT, name VARCHAR(30))	1 row(s) affected, 1 warning(s): 1007 Can't create database 'joins_db'; database exists
4	19:49:03	USE joins_db	0 row(s) affected
5	19:49:03	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
6	19:49:04	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected

STEP-3 INSERT DATA

The screenshot shows the SQL Developer interface with the following SQL statements in the editor:

```
5 DROP TABLE IF EXISTS class_info;
6
7 CREATE TABLE class (
8     id INT,
9     name VARCHAR(30)
10 );
11
12 CREATE TABLE class_info (
13     id INT,
14     address VARCHAR(30)
15 );
16 INSERT INTO class VALUES
17 (1,'Vaishali'),
18 (2,'Vaishali'),
19 (4,'alex');
20 INSERT INTO class_info VALUES
21 (1,'DELHI'),
22 (2,'MUMBAI'),
23 (3,'CHENNAI');
```

The Output window displays the following results:

#	Time	Action	Message
✓ 12	19:51:26	DROP TABLE IF EXISTS class	0 row(s) affected
✓ 13	19:51:26	DROP TABLE IF EXISTS class_info	0 row(s) affected
✓ 14	19:51:26	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
✓ 15	19:51:26	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
✓ 16	19:51:26	INSERT INTO class VALUES (1,'Vaishali'), (2,'Vaishali'), (4,'alex')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 17	19:51:26	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

STEP-4 CROSS JOIN

The screenshot shows the SQL Developer interface with the following SQL statements in the editor:

```
18 (2,'Vaishali'),
19 (4,'alex');
20 INSERT INTO class_info VALUES
21 (1,'DELHI'),
22 (2,'MUMBAI'),
23 (3,'CHENNAI');
24 SELECT *
25 FROM class
26 CROSS JOIN class_info;
```

The Result Grid displays the following data:

	id	name	id	address
▶ 4	alex	1	DELHI	
2	Vaishali	1	DELHI	
1	Vaishali	1	DELHI	
4	alex	2	MUMBAI	
2	Vaishali	2	MUMBAI	
1	Vaishali	2	MUMBAI	
4	alex	3	CHENNAI	
2	Vaishali	3	CHENNAI	
1	Vaishali	3	CHENNAI	

The Output window displays the following results:

#	Time	Action	Message
✓ 13	19:51:26	DROP TABLE IF EXISTS class_info	0 row(s) affected
✓ 14	19:51:26	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
✓ 15	19:51:26	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
✓ 16	19:51:26	INSERT INTO class VALUES (1,'Vaishali'), (2,'Vaishali'), (4,'alex')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 17	19:51:26	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 18	19:52:30	SELECT * FROM class CROSS JOIN class_info	9 row(s) returned

STEP-5 RESET TABLES

Automatic context help is disabled. Use the caret position or to t

```

20 • INSERT INTO class_info VALUES
21   (1,'DELHI'),
22   (2,'MUMBAI'),
23   (3,'CHENNAI');
24 • SELECT *
25   FROM class
26   CROSS JOIN class_info;
27 • DROP TABLE IF EXISTS class;
28 • DROP TABLE IF EXISTS class_info;

```

	id	name	id	address
▶	4	alex	1	DELHI
2	Vaishali	1	DELHI	
1	Vaishali	1	DELHI	
4	alex	2	MUMBAI	
2	Vaishali	2	MUMBAI	
1	Vaishali	2	MUMBAI	
4	alex	3	CHENNAI	
2	Vaishali	3	CHENNAI	
1	Vaishali	3	CHENNAI	

Result 3 x

Output

#	Time	Action	Message
✓	38 19:57:09	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
✓	39 19:57:09	INSERT INTO class VALUES (1,'Vaishali'), (2,'Vaishali'), (4,'alex')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	40 19:57:09	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	41 19:57:09	SELECT * FROM class CROSS JOIN class_info	9 row(s) returned
✓	42 19:57:09	DROP TABLE IF EXISTS class	0 row(s) affected
✓	43 19:57:09	DROP TABLE IF EXISTS class_info	0 row(s) affected

STEP-6 CREATE TABLE AGAIN

Automatic context help is disabled. Use the caret position or to t

```

19   (4,'alex');
20 • INSERT INTO class_info VALUES
21   (1,'DELHI'),
22   (2,'MUMBAI'),
23   (3,'CHENNAI');
24 • SELECT *
25   FROM class
26   CROSS JOIN class_info;
27 • DROP TABLE IF EXISTS class;
28 • DROP TABLE IF EXISTS class_info;
29 • CREATE TABLE class (
30     id INT,
31     name VARCHAR(30)
32   );
33
34 • CREATE TABLE class_info (
35     id INT,
36     address VARCHAR(30)
37   );

```

Output

#	Time	Action	Message
✓	51 19:58:16	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	52 19:58:16	SELECT * FROM class CROSS JOIN class_info	9 row(s) returned
✓	53 19:58:16	DROP TABLE IF EXISTS class	0 row(s) affected
✓	54 19:58:16	DROP TABLE IF EXISTS class_info	0 row(s) affected
✓	55 19:58:16	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
✓	56 19:58:16	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected

STEP-7 INSERT NEW DATA

The screenshot shows the SQL Developer interface with the following components:

- Left Panel:** Contains a tree view with 'GEMENT' (Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, Data Import/Restore), 'NCE' (Startup / Shutdown, Server Logs, Options File), and 'RMANCE' (Dashboard, Performance Reports, Performance Schema Setup). The 'Schemas' tab is selected.
- SQL Editor:** Contains the following SQL code:

```
34 CREATE TABLE class_info (  
35     id INT,  
36     address VARCHAR(30)  
37 );  
38 INSERT INTO class VALUES  
39 (1,'Vaishali'),  
40 (2,'Vaishali'),  
41 (3,'alex'),  
42 (4,'anu');
```
- SQL Additions:** A message on the right states: "Automatic context help is disabled. Use the caret position or to t".
- Result Grid:** Displays the results of the executed statements:

#	id	name	id	address
4	alex	1	DELHI	
2	Vaishali	1	DELHI	
1	Vaishali	1	DELHI	
4	alex	2	MUMBAI	
2	Vaishali	2	MUMBAI	
1	Vaishali	2	MUMBAI	
4	alex	3	CHENNAI	
2	Vaishali	3	CHENNAI	
1	Vaishali	3	CHENNAI	
- Output:** A table showing the execution of the statements:

#	Time	Action	Message
66	20:02:59	DROP TABLE IF EXISTS class	0 row(s) affected
67	20:02:59	DROP TABLE IF EXISTS class_info	0 row(s) affected
68	20:02:59	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
69	20:02:59	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
70	20:02:59	INSERT INTO class VALUES (1,'Vaishali'), (2,'Vaishali'), (3,'alex'), (4,'anu')	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
71	20:02:59	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

STEP-8 INNER JOIN

The screenshot shows the SQL Developer interface with the following components:

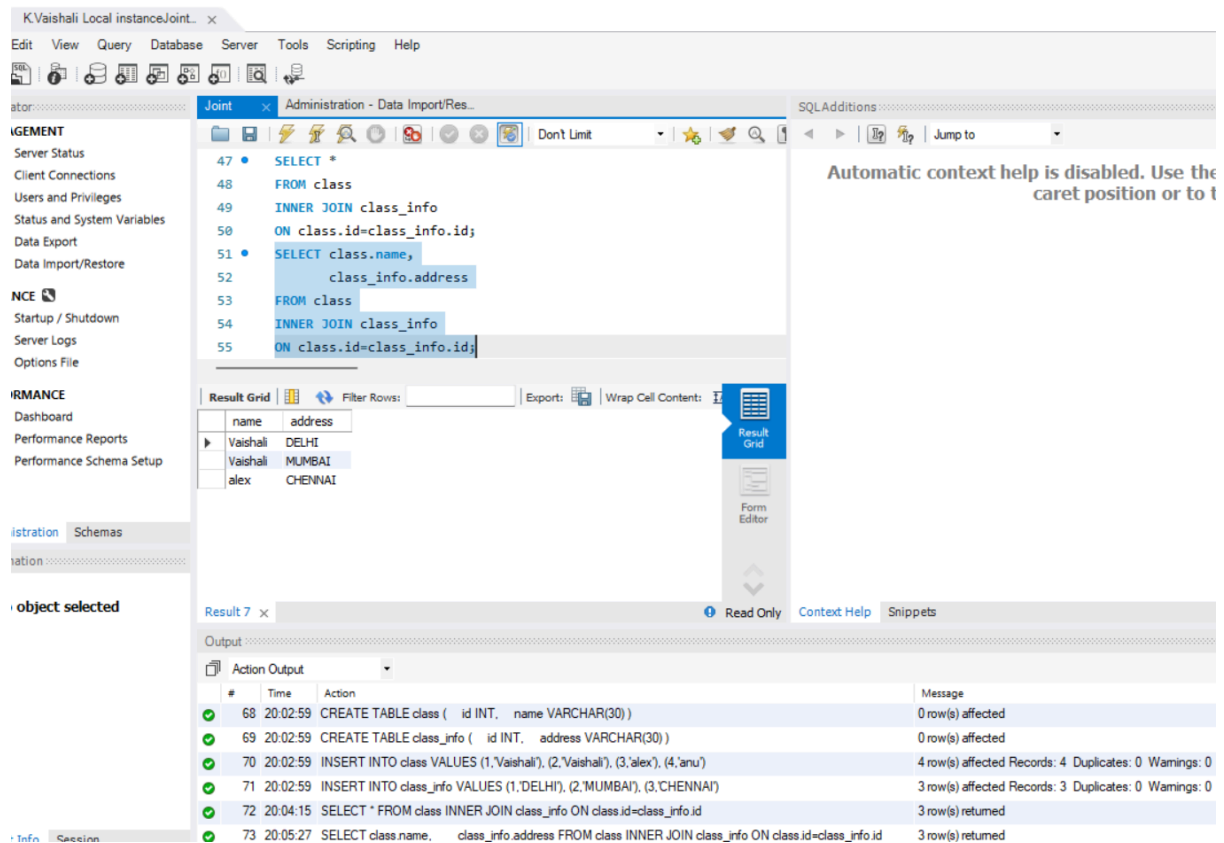
- Left Panel:** Similar to the previous screenshot, with 'Schemas' selected.
- SQL Editor:** Contains the following SQL code:

```
42 (4,'anu');  
43 INSERT INTO class_info VALUES  
44 (1,'DELHI'),  
45 (2,'MUMBAI'),  
46 (3,'CHENNAI');  
47 SELECT *  
48 FROM class  
49 INNER JOIN class_info  
50 ON class.id=class_info.id;
```
- SQL Additions:** A message on the right states: "Automatic context help is disabled. Use the caret position or to t".
- Result Grid:** Displays the results of the executed statements:

#	id	name	id	address
1	Vaishali	1	DELHI	
2	Vaishali	2	MUMBAI	
3	alex	3	CHENNAI	
- Output:** A table showing the execution of the statements:

#	Time	Action	Message
67	20:02:59	DROP TABLE IF EXISTS class_info	0 row(s) affected
68	20:02:59	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
69	20:02:59	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
70	20:02:59	INSERT INTO class VALUES (1,'Vaishali'), (2,'Vaishali'), (3,'alex'), (4,'anu')	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
71	20:02:59	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
72	20:04:15	SELECT * FROM class INNER JOIN class_info ON class.id=class_info.id	3 row(s) returned

STEP-9 INNER JOIN(NAME & ADDRESS)



Automatic context help is disabled. Use the caret position or to t

```
47 SELECT *
48 FROM class
49 INNER JOIN class_info
50 ON class.id=class_info.id;
51 SELECT class.name,
52        class_info.address
53 FROM class
54 INNER JOIN class_info
55 ON class.id=class_info.id;
```

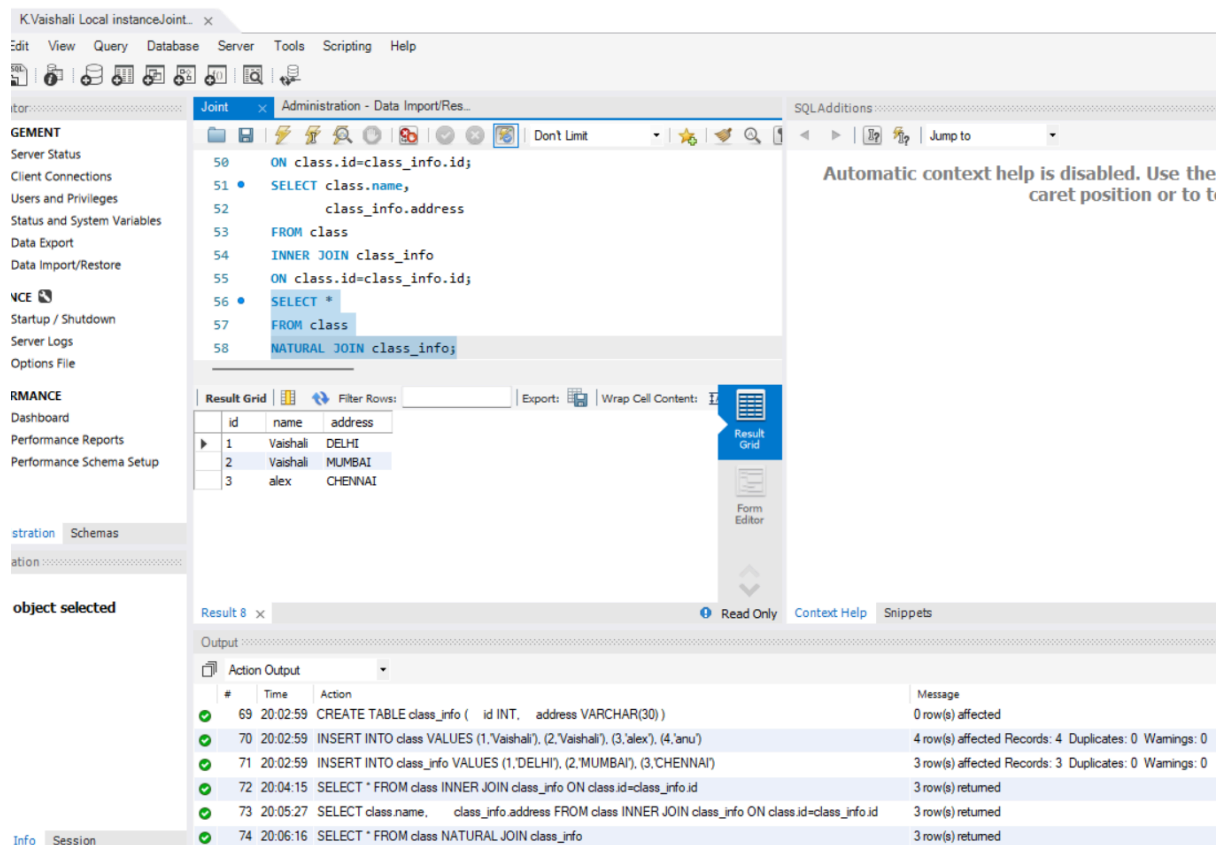
name	address
Vaishali	DELHI
Vaishali	MUMBAI
alex	CHENNAI

Result 7 x

Output

#	Time	Action	Message
68	20:02:59	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
69	20:02:59	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
70	20:02:59	INSERT INTO class VALUES (1,'Vaishali'), (2,'Vaishali'), (3,'alex'), (4,'anu')	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
71	20:02:59	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
72	20:04:15	SELECT * FROM class INNER JOIN class_info ON class.id=class_info.id	3 row(s) returned
73	20:05:27	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id=class_info.id	3 row(s) returned

STEP-10 NATURAL JOIN



Automatic context help is disabled. Use the caret position or to t

```
50 ON class.id=class_info.id;
51 SELECT class.name,
52        class_info.address
53 FROM class
54 INNER JOIN class_info
55 ON class.id=class_info.id;
56 SELECT *
57 FROM class
58 NATURAL JOIN class_info;
```

id	name	address
1	Vaishali	DELHI
2	Vaishali	MUMBAI
3	alex	CHENNAI

Result 8 x

Output

#	Time	Action	Message
69	20:02:59	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
70	20:02:59	INSERT INTO class VALUES (1,'Vaishali'), (2,'Vaishali'), (3,'alex'), (4,'anu')	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
71	20:02:59	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
72	20:04:15	SELECT * FROM class INNER JOIN class_info ON class.id=class_info.id	3 row(s) returned
73	20:05:27	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id=class_info.id	3 row(s) returned
74	20:06:16	SELECT * FROM class NATURAL JOIN class_info	3 row(s) returned

STEP-11 INSERT EXTRA DATA

The screenshot shows the SQL Enterprise Manager interface with the 'Administration - Data Import/Restore' window open. The SQL script in the editor includes several INSERT statements and a SELECT statement. The output pane shows the execution results, including the number of rows affected and the number of records returned.

SQL Script:

```

45 (2,'MUMBAI'),
46 (3,'CHENNAI');
47 • SELECT *
48 FROM class
49 INNER JOIN class_info
50 ON class.id=class_info.id;
51 • SELECT class.name,
52       class_info.address
53 FROM class
54 INNER JOIN class_info
55 ON class.id=class_info.id;
56 • SELECT *
57 FROM class
58 NATURAL JOIN class_info;
59 • INSERT INTO class VALUES
60 (5,'Vaishali');
61 • INSERT INTO class_info VALUES
62 (7,'NOIDA'),
63 (8,'PANIPAT');

```

Output:

#	Time	Action	Message
91	20:07:10	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id=class_info.id	3 row(s) returned
92	20:07:10	SELECT * FROM class NATURAL JOIN class_info	3 row(s) returned
93	20:07:10	INSERT INTO class VALUES (5,'Vaishali')	1 row(s) affected
94	20:07:10	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
95	20:07:32	INSERT INTO class VALUES (5,'Vaishali')	1 row(s) affected
96	20:07:33	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0

STEP-12 LEFT JOIN

The screenshot shows the SQL Enterprise Manager interface with the 'Administration - Data Import/Restore' window open. The SQL script in the editor includes several INSERT statements and a LEFT JOIN query. The output pane shows the execution results, including the number of rows affected and the number of records returned. A result grid is also displayed, showing the data returned by the LEFT JOIN query.

SQL Script:

```

59 • INSERT INTO class VALUES
60 (5,'Vaishali');
61 • INSERT INTO class_info VALUES
62 (7,'NOIDA'),
63 (8,'PANIPAT');
64 • SELECT *
65 FROM class
66 LEFT JOIN class_info
67 ON class.id=class_info.id;

```

Result Grid:

	id	name	id	address
1	Vaishali	1	DELHI	
2	Vaishali	2	MUMBAI	
3	alex	3	CHENNAI	
4	anu		NOIDA	
5	Vaishali		PANIPAT	
6	Vaishali		PANIPAT	

Output:

#	Time	Action	Message
92	20:07:10	SELECT * FROM class NATURAL JOIN class_info	3 row(s) returned
93	20:07:10	INSERT INTO class VALUES (5,'Vaishali')	1 row(s) affected
94	20:07:10	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
95	20:07:32	INSERT INTO class VALUES (5,'Vaishali')	1 row(s) affected
96	20:07:33	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
97	20:09:19	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id	6 row(s) returned

STEP-13 LEFT JOIN WITH NULL

K.Vaishali Local instanceJoint...

File Edit View Query Database Server Tools Scripting Help

Administration - Data Import/Res...

SQLAdditions

Automatic context help is disabled. Use the caret position or to l

```

64 SELECT *
65 FROM class
66 LEFT JOIN class_info
67 ON class.id=class_info.id;
68 SELECT *
69 FROM class
70 LEFT JOIN class_info
71 ON class.id=class_info.id
72 WHERE class_info.id IS NULL;

```

Result Grid

id	name	id	address
4	anu	NULL	NULL
5	Vaishali	NULL	NULL
5	Vaishali	NULL	NULL

Result 14 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message
93	20:07:10	INSERT INTO class VALUES (5,'Vaishali')	1 row(s) affected
94	20:07:10	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
95	20:07:32	INSERT INTO class VALUES (5,'Vaishali')	1 row(s) affected
96	20:07:33	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
97	20:09:19	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id	6 row(s) returned
98	20:10:30	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id WHERE class_info.id IS NULL	3 row(s) returned

Info Session

STEP-14 RIGHT JOIN

K.Vaishali Local instanceJoint...

File Edit View Query Database Server Tools Scripting Help

Administration - Data Import/Res...

SQLAdditions

Automatic context help is disabled. Use the caret position or to l

```

68 SELECT *
69 FROM class
70 LEFT JOIN class_info
71 ON class.id=class_info.id
72 WHERE class_info.id IS NULL;
73 SELECT *
74 FROM class
75 RIGHT JOIN class_info
76 ON class.id=class_info.id;

```

Result Grid

id	name	id	address
1	Vaishali	1	DELHI
2	Vaishali	2	MUMBAI
3	alex	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

Result 15 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message
94	20:07:10	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
95	20:07:32	INSERT INTO class VALUES (5,'Vaishali')	1 row(s) affected
96	20:07:33	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
97	20:09:19	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id	6 row(s) returned
98	20:10:30	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id WHERE class_info.id IS NULL	3 row(s) returned
99	20:11:25	SELECT * FROM class RIGHT JOIN class_info ON class.id=class_info.id	7 row(s) returned

Info Session

STEP-15 RIGHT JOIN WITH NULL

K.Vaishali Local instanceJoint...

File Edit View Query Database Server Tools Scripting Help

Administration - Data Import/Restore

SQLAdditions

Automatic context help is disabled. Use the caret position or to l

```

73 SELECT *
74 FROM class
75 RIGHT JOIN class_info
76 ON class.id=class_info.id;
77 SELECT *
78 FROM class
79 RIGHT JOIN class_info
80 ON class.id=class_info.id
81 WHERE class.id IS NULL;

```

Result Grid

	id	name	id	address
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Result 16 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message
95	20:07:32	INSERT INTO class VALUES (5, 'INSERT INTO class VALUES	1 row(s) affected
96	20:07:33	INSERT INTO class_info VALUE (5, 'Vaishali')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
97	20:09:19	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id	6 row(s) returned
98	20:10:30	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id WHERE class_info.id IS NULL	3 row(s) returned
99	20:11:25	SELECT * FROM class RIGHT JOIN class_info ON class.id=class_info.id	7 row(s) returned
100	20:12:24	SELECT * FROM class RIGHT JOIN class_info ON class.id=class_info.id WHERE class.id IS NULL	4 row(s) returned

Info Session

STEP-16 FULL OUTER JOIN(MYSQL ALTERNATIVE)

K.Vaishali Local instanceJoint...

File Edit View Query Database Server Tools Scripting Help

Administration - Data Import/Restore

SQLAdditions

Automatic context help is disabled. Use the caret position or to l

```

84 LEFT JOIN class_info
85 ON class.id=class_info.id
86
87 UNION
88
89 SELECT *
90 FROM class
91 RIGHT JOIN class_info
92 ON class.id=class_info.id;

```

Result Grid

	id	name	id	address
1		Vaishali	1	DELHI
2		Vaishali	2	MUMBAI
3		alex	3	CHENNAI
4		anu	NULL	NULL
5		Vaishali	NULL	NULL
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Result 17 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message
96	20:07:33	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
97	20:09:19	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id	6 row(s) returned
98	20:10:30	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id WHERE class_info.id IS NULL	3 row(s) returned
99	20:11:25	SELECT * FROM class RIGHT JOIN class_info ON class.id=class_info.id	7 row(s) returned
100	20:12:24	SELECT * FROM class RIGHT JOIN class_info ON class.id=class_info.id WHERE class.id IS NULL	4 row(s) returned
101	20:23:01	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id UNION SELECT * FROM class RIGH...	7 row(s) returned

Info Session

STEP-17 UNMATCHED RECORDS(MYSQL)

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
WHERE class_info.id IS NULL
UNION
SELECT *
FROM class
RIGHT JOIN class_info
ON class.id=class_info.id
WHERE class.id IS NULL;
```

The Results tab displays a table with 4 columns: id, name, id, and address. The data is as follows:

id	name	id	address
4	anu	NULL	NULL
5	Vaishali	NULL	NULL
7	NOIDA	NULL	NULL
8	PANIPAT	NULL	NULL

The Output tab shows the execution log with the following messages:

#	Time	Action	Message
97	20:09:19	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id	6 row(s) returned
98	20:10:30	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id WHERE class_info.id IS NULL	3 row(s) returned
99	20:11:25	SELECT * FROM class RIGHT JOIN class_info ON class.id=class_info.id	7 row(s) returned
100	20:12:24	SELECT * FROM class RIGHT JOIN class_info ON class.id=class_info.id WHERE class.id IS NULL	4 row(s) returned
101	20:23:01	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id UNION SELECT * FROM class RIGH...	7 row(s) returned
102	20:24:38	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id WHERE class_info.id IS NULL UNIO...	4 row(s) returned

STEP-18 CREATE NEW TABLES

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
ON class.id=class_info.id
WHERE class_info.id IS NULL
UNION
SELECT *
FROM class
RIGHT JOIN class_info
ON class.id=class_info.id
WHERE class.id IS NULL;
CREATE TABLE first_table(
  id INT,
  name VARCHAR(30)
);
CREATE TABLE second_table(
  id INT,
  name VARCHAR(30)
);
```

The Output tab shows the execution log with the following messages:

#	Time	Action	Message
99	20:11:25	SELECT * FROM class RIGHT JOIN class_info ON class.id=class_info.id	7 row(s) returned
100	20:12:24	SELECT * FROM class RIGHT JOIN class_info ON class.id=class_info.id WHERE class.id IS NULL	4 row(s) returned
101	20:23:01	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id UNION SELECT * FROM class RIGH...	7 row(s) returned
102	20:24:38	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id WHERE class_info.id IS NULL UNIO...	4 row(s) returned
103	20:25:45	CREATE TABLE first_table(id INT, name VARCHAR(30))	0 row(s) affected
104	20:25:45	CREATE TABLE second_table(id INT, name VARCHAR(30))	0 row(s) affected

STEP-19 INSERT DATA

K:\Vaishali Local instance\Joint_ x

File Edit View Query Database Server Tools Scripting Help

Administration - Data Import/Restore

SQLAdditions: ...

Automatic context help is disabled. Use the caret position or to

```

109      id INT,
110      name VARCHAR(30)
111    );
112
113    CREATE TABLE second_table(
114      id INT,
115      name VARCHAR(30)
116    );
117

```

Result Grid

id	name	id	address
4	anu		
5	Vaishali		
		7	NOIDA
		8	PANIPAT

object selected

Output

Action Output

#	Time	Action	Message
✓ 185	20:31:25	DROP TABLE IF EXISTS first_table	0 row(s) affected
✓ 186	20:31:25	DROP TABLE IF EXISTS second_table	0 row(s) affected
✓ 187	20:31:25	CREATE TABLE first_table(id INT, name VARCHAR(30))	0 row(s) affected
✓ 188	20:31:25	CREATE TABLE second_table(id INT, name VARCHAR(30))	0 row(s) affected
✓ 189	20:31:25	INSERT INTO first_table VALUES (1,'Vaishali'), (2,'Vaishali')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓ 190	20:31:25	INSERT INTO second_table VALUES (2,'adam'), (3,'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0

Info Session

STEP-20 UNION

K:\Vaishali Local instance\Joint_ x

File Edit View Query Database Server Tools Scripting Help

Administration - Data Import/Restore

SQLAdditions: ...

Automatic context help is disabled. Use the caret position or to

```

115      name VARCHAR(30)
116    );
117    SELECT *
118    FROM first_table
119
120    UNION
121
122    SELECT *
123    FROM second_table;

```

Result Grid

id	name
1	Vaishali
2	adam
3	chester

object selected

Output

Action Output

#	Time	Action	Message
✓ 186	20:31:25	DROP TABLE IF EXISTS second_table	0 row(s) affected
✓ 187	20:31:25	CREATE TABLE first_table(id INT, name VARCHAR(30))	0 row(s) affected
✓ 188	20:31:25	CREATE TABLE second_table(id INT, name VARCHAR(30))	0 row(s) affected
✓ 189	20:31:25	INSERT INTO first_table VALUES (1,'Vaishali'), (2,'Vaishali')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓ 190	20:31:25	INSERT INTO second_table VALUES (2,'adam'), (3,'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓ 191	20:32:58	SELECT * FROM first_table UNION SELECT * FROM second_table	4 row(s) returned

Info Session

STEP-21 UNION(NAMES)

K.Vaishali Local instanceJoint...

Edit View Query Database Server Tools Scripting Help

Administration - Data Import/Restore

SQL Additions

Automatic context help is disabled. Use the caret position or to

```

122 SELECT *
123 FROM second_table;
124 SELECT name
125 FROM first_table
126
127 UNION
128
129 SELECT name
130 FROM second_table;

```

Result Grid

name
Vaishali
Vaishali
adam
chester

object selected

Result 50

Output

#	Time	Action	Message
187	20:31:25	CREATE TABLE first_table(id INT, name VARCHAR(30))	0 row(s) affected
188	20:31:25	CREATE TABLE second_table(id INT, name VARCHAR(30))	0 row(s) affected
189	20:31:25	INSERT INTO first_table VALUES (1,'Vaishali'), (2,'Vaishali')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
190	20:31:25	INSERT INTO second_table VALUES (2,'adam'), (3,'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
191	20:32:58	SELECT * FROM first_table UNION SELECT * FROM second_table	4 row(s) returned
192	20:34:29	SELECT name FROM first_table UNION SELECT name FROM second_table	4 row(s) returned

STEP-22 UNION ALL

K.Vaishali Local instanceJoint...

Edit View Query Database Server Tools Scripting Help

Administration - Data Import/Restore

SQL Additions

Automatic context help is disabled. Use the caret position or to

```

129 SELECT name
130 FROM second_table;
131 SELECT *
132 FROM first_table
133
134 UNION ALL
135
136 SELECT *
137 FROM second_table;

```

Result Grid

id	name
1	Vaishali
2	Vaishali
2	adam
3	chester

object selected

Result 51

Output

#	Time	Action	Message
188	20:31:25	CREATE TABLE second_table(id INT, name VARCHAR(30))	0 row(s) affected
189	20:31:25	INSERT INTO first_table VALUES (1,'Vaishali'), (2,'Vaishali')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
190	20:31:25	INSERT INTO second_table VALUES (2,'adam'), (3,'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
191	20:32:58	SELECT * FROM first_table UNION SELECT * FROM second_table	4 row(s) returned
192	20:34:29	SELECT name FROM first_table UNION SELECT name FROM second_table	4 row(s) returned
193	20:35:29	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned

